

END SEMESTER EXAM

①

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \frac{2n^2+1}{n^2+3n} = 2, \text{ Since } \frac{2n^2+1}{n^2+3n} = \frac{2 + \frac{1}{n^2}}{1 + \frac{3}{n}} \rightarrow 2 \text{ as } n \rightarrow \infty$$

now we verify $\lim_{n \rightarrow \infty} \frac{2n^2+1}{n^2+3n} = 2$.

$$\left| \frac{2n^2+1}{n^2+3n} - 2 \right| = \left| \frac{2n^2+1-2n^2-6n}{n^2+3n} \right| = \frac{6n-1}{n^2+3n} < \frac{6n}{n^2} = \frac{6}{n} < \epsilon$$

whenever $n > \frac{6}{\epsilon}$. [Since $6n-1 < 6n$
and $n^2+3n > n^2$
 $\forall n$]

choose $N = \left[\frac{6}{\epsilon} \right] + 1$.

So, $\left| \frac{2n^2+1}{n^2+3n} - 2 \right| < \epsilon$ for all $n \geq N$, where $N = \left[\frac{6}{\epsilon} \right] + 1$.

②

$$x_{n+1} - x_n = \frac{6(1+x_n)}{7+x_n} - x_n = \frac{6 - x_n - x_n^2}{7+x_n} = \frac{(2-x_n)(3+x_n)}{7+x_n}$$

$$\text{and } 2 - x_{n+1} = 2 - \frac{6(1+x_n)}{7+x_n} = \frac{4(2-x_n)}{7+x_n}.$$

Case 1: $x_n < 2$: If $x_n < 2$, then $x_{n+1} > x_n$ and $x_{n+1} < 2 \forall n$.

i.e., if $x_n < 2$, then $x_n < x_{n+1} < 2 \forall n$.

So, if $x_1 = c < 2$, $x_1 < x_2 < x_3 < \dots < 2$, i.e., the sequence $\{x_n\}$ is monotone increasing when $c < 2$.

now the sequence $\{x_n\}$ is bounded above, where its upper bound is 2, i.e., $x_n < 2 \forall n$.

The sequence $\{x_n\}$ is monotone increasing and bounded above and hence it converges.

Case 2: $x_n > 2$: Similarly, if $x_1 = c > 2$, $x_1 > x_2 > x_3 > \dots > 2$

and the sequence $\{x_n\}$ is monotone decreasing and bounded below, where its lower bound is 2. i.e., $x_n > 2 \forall n$.

So the sequence is monotone decreasing and bounded below and hence it converges.

Since the sequence $\{x_n\}$ is convergent, let $\lim_{n \rightarrow \infty} x_n = l$.

$$x_{n+1} = \frac{6(1+x_n)}{7+x_n}$$

Taking limit —

$$\lim_{n \rightarrow \infty} x_{n+1} = \frac{6(1 + \lim_{n \rightarrow \infty} x_n)}{(7 + \lim_{n \rightarrow \infty} x_n)} \Rightarrow l = \frac{6(1+l)}{7+l}$$

$$\Rightarrow l^2 + l - 6 = 0$$

$$(l+3)(l-2) = 0$$

$$\Rightarrow l = -3, \text{ or } 2.$$

But the terms of the sequence are all positive and hence the limit cannot be negative.

So the limit is 2.

③ (i) $u_1 = 1, u_{n+1} = \frac{(-1)^n}{2} \left\{ |u_n| + \frac{2}{|u_n|} \right\}, n \geq 1$

$$u_{n+1} = -\frac{1}{2} \left\{ |u_n| + \frac{2}{|u_n|} \right\} \text{ if } n \text{ is odd}$$

$$= \frac{1}{2} \left\{ |u_n| + \frac{2}{|u_n|} \right\} \text{ if } n \text{ is even}$$

$$u_1 = 1, u_2 = -1.5, u_3 = 1.42, u_4 = -1.41, u_5 = 1.41$$

$$u_6 = -1.41, u_7 = 1.41$$

now take subsequence $\{u_{2n-1}\} = \{u_1, u_3, u_5, u_7, \dots\}$
 $= \{1, 1.42, 1.41, 1.41, \dots\}$

This subsequence is clearly convergent and converges to 1.41

now take another subsequence $\{u_{2n}\} = \{u_2, u_4, u_6, u_8, \dots\}$
 $= \{-1.5, -1.41, -1.41, -1.41, \dots\}$

This subsequence is also convergent and converges to -1.41.

now we have to get the actual limit of the subsequences.

Let $\lim_{n \rightarrow \infty} u_{2n-1} = l$ and $\lim_{n \rightarrow \infty} u_{2n} = -l$.

$$u_{n+1} = \frac{(-1)^n}{2} \left\{ |u_n| + \frac{2}{|u_n|} \right\}$$

$$\therefore u_{2n} = \frac{(-1)^{2n-1}}{2} \left\{ |u_{2n-1}| + \frac{2}{|u_{2n-1}|} \right\}$$

$$\therefore \lim_{n \rightarrow \infty} u_{2n} = -\frac{1}{2} \left\{ \lim_{n \rightarrow \infty} |u_{2n-1}| + \frac{2}{\lim_{n \rightarrow \infty} |u_{2n-1}|} \right\}$$

$$\therefore -l = -\frac{1}{2} \left\{ l + \frac{2}{l} \right\}$$

$$\therefore l = \frac{1}{2} \left(l + \frac{2}{l} \right) \text{ as } 2l = l + \frac{2}{l}$$

$$\text{as } l = \pm \sqrt{2}.$$

$$\boxed{\lim u_n = \sqrt{2} \text{ and } \lim u_n = -\sqrt{2}}$$

③ (ii) Let $u_n = \frac{(2n+1)(2n+2) \dots (2n+n)}{n^n}, n \geq 1$

$$u_{n+1} = \frac{(2n+3)(2n+4) \dots (2n+n)(2n+n+1)(2n+n+2)(2n+n+3)}{(n+1)^{n+1}}$$

$$\begin{aligned} \frac{u_{n+1}}{u_n} &= \frac{(3n+1)(3n+2)(3n+3)}{(2n+1)(2n+2)(n+1)^{n+1}} \cdot n^n = \frac{(3n+1)(3n+2)(3n+3)}{(2n+1)(2n+2)(n+1)} \left(\frac{n}{n+1} \right)^n \\ &= \frac{(3+\frac{1}{n})(3+\frac{2}{n})(3+\frac{3}{n})}{(2+\frac{1}{n})(2+\frac{2}{n})(1+\frac{1}{n})} \frac{1}{(1+\frac{1}{n})^n} \end{aligned}$$

$$\text{now } \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(3+\frac{1}{n})(3+\frac{2}{n})(3+\frac{3}{n})}{(1+\frac{1}{n})(2+\frac{1}{n})(2+\frac{2}{n})} \cdot \frac{1}{\lim_{n \rightarrow \infty} (1+\frac{1}{n})^n}$$

$$= \frac{27}{4e} \quad \left[\text{since } \lim_{n \rightarrow \infty} (1+\frac{1}{n})^n = e \right]$$

now by Cauchy's second limit theorem

$$\lim_{n \rightarrow \infty} u_n^{\frac{1}{n}} = \frac{27}{4e} = \lim_{n \rightarrow \infty} x_n, \text{ limit is } \frac{27}{4e}.$$

④ (i) $b_n = 2 - \frac{(-1)^n}{n}, a_n = b_n - b_{n+1}, \forall n.$

$$a_n = b_n - b_{n+1} = 2 - \frac{(-1)^n}{n} - 2 + \frac{(-1)^{n+1}}{n+1}$$

$$= \frac{(-1)^{n+1}}{n+1} - \frac{(-1)^n}{n} = (-1)^{n+1} \left[\frac{1}{n} + \frac{1}{n+1} \right]$$

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k = \lim_{n \rightarrow \infty} S_n, \text{ where } \{S_n\} \text{ is the sequence of } n^{\text{th}} \text{ partial sum.}$$

$$S_n = \sum_{k=1}^n (-1)^{k+1} \left[\frac{1}{k} + \frac{1}{k+1} \right]$$

$$= \left(1 + \frac{1}{2}\right) - \left(\frac{1}{2} + \frac{1}{3}\right) + \left(\frac{1}{3} + \frac{1}{4}\right) - \left(\frac{1}{4} + \frac{1}{5}\right) + \dots$$

and $+ \left(\frac{1}{n} + \frac{1}{n+1}\right), \text{ when } n \text{ is odd}$

$$= \left(1 + \frac{1}{2}\right) - \left(\frac{1}{2} + \frac{1}{3}\right) + \left(\frac{1}{3} + \frac{1}{4}\right) - \left(\frac{1}{4} + \frac{1}{5}\right) + \dots$$

$- \left(\frac{1}{n} + \frac{1}{n+1}\right), \text{ when } n \text{ is even.}$

when n is odd, $S_n = 1 + \frac{1}{n+1} \Rightarrow \lim_{n \rightarrow \infty} S_n = 1.$

when n is even, $S_n = 1 - \frac{1}{n+1} \Rightarrow \lim_{n \rightarrow \infty} S_n = 1.$

one can find it by Leibnitz test

So the series is convergent.

④ (ii) $\left(\frac{1}{2 \cdot 3} + \frac{1}{2^2 \cdot 3}\right) + \left(\frac{1}{2^2 \cdot 3^2} + \frac{1}{2^3 \cdot 3^2}\right) + \left(\frac{1}{2^3 \cdot 3^3} + \frac{1}{2^4 \cdot 3^3}\right) + \dots$

$$= \sum_{n=1}^{\infty} \left[\frac{1}{2^n 3^n} + \frac{1}{2^{n+1} 3^n} \right]. \text{ let } x_n = \frac{1}{2^n 3^n} + \frac{1}{2^{n+1} 3^n}$$

$$= \frac{1}{2^n 3^n} \left[1 + \frac{1}{2} \right] = \frac{3}{2} \cdot \frac{1}{6^n}.$$

$$\sum_{n=1}^{\infty} x_n = \sum_{n=1}^{\infty} \frac{3}{2} \cdot \frac{1}{6^n} = \frac{3}{2} \left(\sum_{n=1}^{\infty} \frac{1}{6^n} \right) = \frac{3}{2} \cdot \frac{\frac{1}{6}}{1 - \frac{1}{6}} = \frac{3}{10}$$

geometric series with common ratio < 1 .

So the series sum is $\frac{3}{10}.$

⑤ (i) If $\sum_{n=1}^{\infty} a_n$ Converges, $\lim_{n \rightarrow \infty} a_n = 0$ [My theorem]

$$\lim_{n \rightarrow \infty} \frac{a_n - 1}{n^n} = - \lim_{n \rightarrow \infty} \frac{1}{n^n} = -1$$

[Since $\lim_{n \rightarrow \infty} a_n = 0$
and $\lim_{n \rightarrow \infty} n^n = \infty$]

⑤ (ii) Let $\sum_{n=1}^{\infty} u_n$, where $u_n = \frac{1^2 \cdot 3^2 \cdot 5^2 \dots (2n-1)^2}{2^2 \cdot 4^2 \cdot 6^2 \dots (2n)^2} x^{n-1}$

Since $x > 0$, each term of the series is positive
i.e., $\sum_{n=1}^{\infty} u_n$ is a positive series.

$$u_{n+1} = \frac{1^2 \cdot 3^2 \cdot 5^2 \dots (2n-1)^2 (2n+1)^2}{2^2 \cdot 4^2 \cdot 6^2 \dots (2n)^2 (2n+2)^2} \cdot x^n$$

$$\frac{u_{n+1}}{u_n} = \frac{(2n+1)^2}{(2n+2)^2} x$$

$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = x$. Now by applying D'Alembert's Ratio Test

$\sum u_n$ Converges if $x < 1$, i.e. if $0 < x < 1$
and diverges if $x > 1$, i.e. if $1 < x < \infty$.

⑥ $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^3 + a}$, Let $x_n = \frac{n}{n^3 + a}$

$\sum x_n$ is absolutely Convergent if $\sum |x_n|$ is Convergent.

$\sum x_n$ is Conditionally Convergent if $\sum x_n$ Converges but $\sum |x_n|$ does not Converge.

Now check $\sum_{n=1}^{\infty} \left| (-1)^{n-1} \cdot \frac{n}{n^3 + a} \right|$ is Convergent or not.

i.e. $\sum_{n=1}^{\infty} \frac{n}{n^3 + a}$ is Convergent or not.

$$x_n = \frac{n}{n^3 + a}, n \geq 1, \text{ Take } y_n = \frac{1}{n^2}, n \geq 1$$

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \lim_{n \rightarrow \infty} \frac{n^2}{n^3 + a} = 1 < \infty.$$

$\sum_{n=1}^{\infty} y_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ is Convergent by p-series $\sum \frac{1}{n^p}$ for $p > 1$.

By applying limit comparison test,

Since $\sum y_n$ is Convergent, $\sum_{n=1}^{\infty} \frac{n}{n^3 + a}$ is Convergent

So the given series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^3 + a}$ is absolutely Convergent.

⑦ (i) Take $\{x_n\}$ where $x_n = 2n\pi$
 $x_n \rightarrow \infty$ as $n \rightarrow \infty$. $f(x_n) = x_n^2 \operatorname{sgn} \{ \cos x_n \}$
 $= (2n\pi)^2 \operatorname{sgn} \{ \cos 2n\pi \}$

So $x_n \rightarrow \infty$ as $n \rightarrow \infty \Rightarrow f(x_n) \rightarrow \infty$ as $n \rightarrow \infty$.
 Let $y_n = \pi + 2n\pi$,
 $y_n \rightarrow \infty$ as $n \rightarrow \infty$

$$\begin{aligned} f(y_n) &= y_n^2 \operatorname{sgn}(\cos y_n) = y_n^2 \operatorname{sgn}(\cos(\pi + 2n\pi)) \\ &= (\pi + 2n\pi)^2 \operatorname{sgn}(-\cos 2n\pi) \\ &= -(\pi + 2n\pi)^2 \rightarrow -\infty, \text{ as } n \rightarrow \infty \end{aligned}$$

So $y_n \rightarrow \infty$ as $n \rightarrow \infty \Rightarrow f(y_n) \rightarrow -\infty$ as $n \rightarrow \infty$

Hence we have two sequences $\{x_n\}$ and $\{y_n\}$ both diverging to ∞ but the sequence $\{f(x_n)\}$ and $\{f(y_n)\}$ approach to two different limits.

By sequence criterion, $\lim_{x \rightarrow \infty} x^2 \operatorname{sgn}(\cos x)$ does not exist.

(ii) $f(x) = \begin{cases} x^2, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x)}{x}$$

We know that $0 \leq \frac{f(x)}{x} \leq x$, for $x > 0$

and $x \leq \frac{f(x)}{x} \leq 0$ for $x < 0$.

By Sandwich theorem $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0$. Since $\lim_{x \rightarrow 0} x = 0$

So $f'(0)$ exist and it is equal to 0.

f is differentiable at 0.

⑧ (i) $f(x) = e^x \sin x + 1$
 Let a and b be any two real roots of $f(x) = 0$ and $a < b$.
 Then $f(a) = f(b) = 0$.

$f(x)$ is continuous on $[a, b]$ and differentiable in (a, b) .

So by Rolle's theorem $\exists c \in (a, b)$ such that $f'(c) = 0$.

$$\text{So } f'(c) = 0 \Rightarrow e^c \sin c + e^c \cos c = 0$$

$$\text{or, } \tan c + 1 = 0$$

$\Rightarrow c$ is a real root of the eqⁿ $\tan x + 1 = 0$ where
 $c \in (a, b)$.

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⑧(ii) Since $f(x)$ is polynomial function defined in $[1, 2]$, the function $f(x)$ is continuous on $[1, 2]$ and differentiable in $(1, 2)$.

So $f'(x) = 2x + 3$ exists in $(1, 2)$.

Hence MVT is applicable.

Using this theorem, $\exists c \in (1, 2)$,

$$f'(c) = \frac{f(2) - f(1)}{2 - 1} = 12 - 6 = 6.$$

$$\therefore 2c + 3 = 6 \Rightarrow \boxed{c = \frac{3}{2}}$$

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